

no. 15 Language large language Model

It is a type of AI built or trained on deep neural networks designed to process, understand and generate human like text.

How it works on high level

- It is able to take thousands of GPUs (Graphic Processing unit) running for months to train.
- It is trained on trillions of words from books, websites, articles, code.

Difference between MLP and LLM

Aspect	LLM	MLP
Scope	Subcategory within MLP	Broad field of AI
Data needs	Extremely high	Moderate
Performance	Superior for complex language understanding - Reason through steps if prompt well.	Efficient for specific tasks.

Two well known LLMs

a) GPT-4 / GPT-5 (by open AI) ~~chatGPT~~ chatGPT relies or functions on LLM's. It has a very strong reasoning, coding used for ~~Microsoft~~ Microsoft copilot

- AI features in Gmail/docs, Android

- Gemini (by google deep mind) known for multimodel.
Handles text, images, audio. used in it is used in :-
- google Gemini App
 - AI features in Gmail / Docs, Android

10.16 Generative AI - META AI

author - model, mase spark, year - 2026

Generative AI - it is a tool which creates new content, instead of just analysing or classifying existing stuff. It draws millions of examples to learn patterns of that content type. and when you prompt it predicts what example should look like piece by piece.

How Generative AI works at high level.

a) Training - learns patterns
- feed complex data millions or billions it doesn't memorise them, it tweaks billions of internal parameters to learn learn.

b) Architecture - transforms for most modern ones. The key trick here is "attention" for each piece it is generating, It looks back at all previous pieces to decide what fits next.

Generation - predict step by step.

text LLMs - predict the next token / word

example - "a dog is barking - - - - -" → "loudly"

No. 17 Training Dataset

it is a collection of data used to teach a machine learning model how to make predictions, recognise patterns or generate outputs.

■ Quality Training Data - is important because AI can only learn what you feed it.

a) Accuracy depends on it - if training data has wrong facts, bias, or noise the model takes that as "truth"

b) Noisy and messy

■ Scrapped websites websites with ads - mixed in
"buy now! Click here - the capital of France is Paris by one and get 1 free"

■ Duplicate data - same wikipedia article copied 1000x
→ model thinks that text is important.

Toxic or harmful

- comments section full of insults → model learns that aggressive tone is "normal".

Bad output out Bias Data Training

example - hiring data: men hired for tech 90% of the time
output model produces: ranks male-coded resumes higher for software jobs.

Example 2 → Sentiment data only from US Twitter
model thinks → South African slang like ("erish")
only implies negativity

no. 18 Computer Vision

IBM - Authors Rina Diae Caballer & Cole Stryker
- www.ibm.com/think/topics/computer-vision
meta AI

Computer Vision - is subfield of AI that enables machines with the ability to process, analyse and interpret visual inputs such images and videos. It makes use of machine learning, deep learning and neural and neural networks to recognise patterns, objects and scenes.

How do machines "see" and interpret image

1) Image becomes a grid of numbers - photo is broken into pixels to a model "cap photo" starts as a giant list of numbers.

2) Training teaches what patterns mean

it never learns "cat" as concept → it learns : when pixels have this statistical pattern of edges / color / texture output the label "cat".

SECTION C - APIs, TOOLS & DEVELOPER CONCEPTS

www.bing.com/copilot/API

10.19

API - (Interface Programming Interface)
is an application designed to allow applications, platforms, or services to interact with customer related data or functionalities in a ~~secure~~ secure and standardized way.

APIs - allows apps to "borrow" features from other services instead of building everything from scratch.

Exempl-1 - Uber ordering - when you open ~~uber~~ uber & cars on a map

MAPS - uber calls google maps to show map tiles plus your location.

Uber Payment API - when you pay, uber calls Stripe/Paypal to charge your card. Uber never sees your card number.

Example 2 - Instagram login with facebook

- Instagram sends you to facebook's auth API

↳ you approve → facebook API sends instagram a token saying "this user @gmaabb, verified"

↳ Instagram lets you in without user creating a new password.

No. 20 API Key - www.blogpostman.com

API Key - It is a unique string of randomly generated characters used to authenticate clients and grant access to an API.

Purpose of API

- a) Authentication - confirms request is from me not a random user.
- b) Authorisation - checks if your plan allows this action. (checks if you are using free/paid version of the app)
- c) Tracking/Billing - counts how many requests you've made to bill you on or cut you because you've reached limit.

What happens if exposed publicly

1) Unauthorised usage and huge bills

If anyone who stumbles upon your key call API as you e.g. anyone post Open AI key GitHub. A bot scrapes it and runs 10M requests like AWS, Google Cloud, Open AI bill the key owner, not abuser.

2) Abuse for spam/illegal activity

Attackers/hackers use your key for crypto mining, send spam SMS. Whole activity will be tracked back to your account → you are banned/cut off from app usage.

no. 21 REST API : META API

A rest API - is an application programming interface (API) that conforms to the design principles of the representational State transfer. It works by exposing data as resources with URLs, and use HTTP methods to act on them.

REST (Representational State Transfer)

HTTP method	Action	Example
GET	→ read data	→ Get/users/123 → get user 123
POST	→ Create data	→ Post/users → create → create new user
PUT	→ update / replace	→ Put/users/123 → replace user 123
PATCH	→ Partial update	→ Patch/users/123 → update name only
DELETE	→ Remove	→ Delete/users/123 → delete user

no. 22 Json (Java script object notation)

it is a text format for storing and sending data, and its human readable, machine friendly. It is more like C language) most APIs.

Json is commonly used when working with APIs and data exchange because

1) Human - readable + machine - readable

Example - one can glance at Json and understand it. No angle brackets or schemas needed

Json

```
{ "name": "Siya", "age": 24 }
```

XML

```
<person><name>Siya</name><age></person>
```


Json

- is lightweight
- no closing tags; no metadata = faster Requests
- mobile apps on data in pinedown notice difference

SIMPLE JSON OBJECT:

Json

```
{  
  "name": "Siya"  
  "age": 24,  
  "city": "pinedown"  
  "is student": true  
}
```

What's in it

"name": String value

"age": number value

"city": String value

"is student": boolean value true OR false

That's it. Key-value pairs inside { }. This what
an API would send if you did GET/user

Mo. 23 Prompt Engineering

Prompt Engineering = how you ask AI to get better results

- it is a skill of writing instructions for LLMs like GPT-4 or Meta's Claude, so they give you exactly what you want

Importance of Prompt Engineering with Chat GPT / Claude

- Both models are smart, but they're not mind readers, prompt decides if gold or generic fluff.

1) Accuracy - Examples

Expt 1 - poor prompt - "tell me about diabetes"

Results - Wikipedia summary, maybe too broad or too advanced.

Good prompt - "you are unsure - explain type-2 to a 40 year old in pinetown with no medical background."

Use 3 bullet points, avoid jargon, include local diet.

Result - useful answer tailored to the person for easier interpretation

Expt - 2 - Saves money + time

Chat GPT - Plus and Claude Pro, charge by usage.
APIs bill per token.

Bad : 5 back and forths to get right answer = 5x cost

Good : 1 clear prompt API's q 1

Good : 1 clear prompt gets it right first try.

NB : if you use API's good prompts cut your bill
50-80%

3) Reduces Hallucinations

Lm's make stuff up when unsure. Prompt engineering grounds them.

TRICKS : "if you don't know, say 'I don't know'" (if
Source use only info from this text.)

Claude especially follows instructions well
when told to be strict.

no. 24 MAP function

Map function is built in function in programming languages like Python and Javascript that allows to apply a specific function.

functions of Map function in programming

- It takes a list; applies a function to each element and gives you back a new list with results. Original list stays unchanged.

Example - Python

Python:

```
numbers = [1, 2, 3, 4]
```

```
doubled = list(map(lambda x: x * 2, numbers))
```

```
Print (doubled) # [2, 4, 6, 8]
```


Nov 25 Cloud Computing

www.agamamtechnologies.com

Cloud Computing - is the on demand delivery of Computing.

Resources - such as Servers, storage, database, networking, software and analytics - over the internet, with pay as you go pricing model.

Instead of owning physical infrastructures, organisations rent resources from providers like.

3 main Cloud Service Models

1) IaaS - infrastructure as a service.

It is renting the computers instead of buying them. You get virtual servers, storage and networking over the internet. You manage the software, provider manages the hardware.

→ it enables clients/business to scale their IT infrastructure dynamically, pay only what they use and avoid the complexities of maintaining physical hardware. It is a cornerstone of modern cloud computing, supporting innovation and growth in this digital era.

Example - ALIBABA Cloud

It is a major IaaS provider in Asia. Offers services such as virtual machines, storage, networking and database. Suitable for enterprise workloads and big performance computing.

PaaS - Platform as a service

its function is to manage all hardware and software resources to develop applications through cloud. It is "also renting the platform, not just computers".

You write a code and provides handles the servers OS, scaling, runtime and updates.

PaaS Example: Deploying a python REST API without touching server. Lets build the fast API app.

Python

main python

```
from fastAPI import fast API
app = fastAPI()
```

```
@app.get("/")
```

```
def home():
```

```
    return {"message": "API from Pindtown"}
```

Deploying on PaaS like Render

1) Push code to GitHub

2) Go to Render.com → "new website service" →

connect repo

3) Settings:

- Runtime: Python

- build command: pip install → requirements.txt

- Start command: uvicorn main: app --

- host 0.0.0.0 -- port 10000

Click deploy

Results: 2 min later you get `https://your-opion-render.com` with HTTPS, auto-restart, logs, and it scales it up if spikes.
You never saw a server.

3) SaaS - software as a service

It is a software delivery model where applications are hosted by a server service provider and made available to customers over the ~~internet~~ internet. ~~lasted~~
Instead of purchasing and installing software on individual devices, users can access the software through a web browser or mobile App subscription basis.

SaaS Example

- Netflix / Showmax - a streaming service for movies and TV shows.
- Sales force - a customer relationship management (CRM)
- Google workspace - a suit of productivity tools including gmail, google docs and G-Drive.

No. 26 Version Control System (VCS)

www.geeksforgeeks.org/git/version-control-system

VCS - it is a tool used in software development and collaborative projects to track and manage changes to source code. It allows multiple developers to collaborate on the same project without overwriting each other's works.

Git = Version Control for your code.

It is a tool that tracks every change you make to files.

It enables you to save snapshots, go back in time, work with teammates and not lose work.

for instance - "git does version history, but for code and way more powerful."

No. 27 Jupyter Notebook

Jupyter - is a document where you can mix code, results and notes in one place.

It runs in your browser. You write code in "cells" hit Shift + Enter and see output right below.

Jupyter notebooks use python (by mile) Jupyter was built for python originally as "ipython notebook"

- It is popular among Data Analysts and Scientists because their work isn't "write app → ship it"

- Its "explore → test → explain → repeat"

Jupyter fits the loop perfectly.

no. 28 Python

It is programming language that's ~~as~~ easy to read and simple. Created by Guido Van Rossum and was first released in 1991. It supports multiple programming paradigms, including procedural object oriented and functional programming.

WHY DATA ANALYSTS USE IT

It has simple Syntax that is similar to English language which allows developers to write programs with fewer lines of code compared to other programming languages.

WHY AI uses Python

It uses python because it is the ~~best~~ BEST "glue" language for research + Production. AI isn't just a code, it's math + data + experimentation + ~~ex~~ GPUs. python
Python rules all 4

four widely used ~~pt~~ python libraries

1) NumPy

- Purpose: Numerical computing with support for large multidimensional arrays and matrices.
- Key features: High performance mathematical functions and vectorized operations.

2) Pandas

- Purpose - data manipulation and analysis
- Key feature - Data frame series objects for handling structured data efficiently

3) matplotlib

- Purpose - Data visualisation
- Key feature - Create static, animated, and interactive plots in python.

4) Requests

- Purpose : Simplified HTTP requests for APIs and web services.
- Key feature : Easy to use methods for GET, POST, PUT, DELETE, PATCH.

no. 29 Data Privacy

It is also called information privacy - It refers to an individual's right to control the circumstances under which their personal data is handled.

Personal data includes;

— ID such as names, address, email, financial details, health records and online behaviour

Importance

It is meant to ensure that individuals maintain authority over their data privacy build customer trust & loyalty which enhances brand reputation and business.

Legal Compliance - Majority of countries have set laws and regulations such as the GDPR that require organisations to uphold data privacy.

Freedom & autonomy - privacy is considered a fundamental human rights, allowing individuals to control their personal info and maintain autonomy over their digital and real world presence

GDPR (General Data Protection Regulation)

it is EU law that controls how companies collect, store and use people's personal data, came into effect on 25th May 2018. Gives right to allow or prohibit organisation to use it in 2018 and can demand for it to be erased.

it applies to all EU, EEA Countries, UK countries.

it is also applicable to countries which their organisations offer goods and services to EU citizens (meaning US, Canada or Australia must comply with GDPR when processing personal data of EU/EEA/UK citizens)

South Africa recently adopted one similar method effect July 1st 2020 called POPIA. it is designed to protect personal info & aligns closely with GDPR principles including -